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Gaming Monitors

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When contemplating a kickass gaming rig, often the priority given to a graphics card far overshadows everything else – keyboard, mice, headphones, monitor, ergonomic chair, etc. In our haste to gush over the latest and greatest GPU, we mustn't forget that every component is an important piece of the jigsaw which completes your gaming experience puzzle. And picking the right monitor can make or break your visual gaming experience. Especially when monitors are getting more intimately connected to new graphics technologies like NVIDIA G-Sync and AMD FreeSync.

Picking just any monitor off the shelf to satisfy your visual needs may work when you're trying to have something to watch movies on or attend to your non-specialized work needs. Most mainstream monitors are pretty good all-rounders when it comes to being jack of all trades, even encompassing features like USB port extenders, audio speakers, and built-in webcams for those video chat sessions. However, gaming – which often is a performance intensive activity both on the device and user level (you gotta bring your A game to kill enemies and pwn n00bs, don't you?) – needs monitors to have a few specialised skills to ensure they don't drop the ball when you're into the thick of things.

Setting a budget and picking a monitor is one thing, but finding the right monitor tuned to your gaming rig is even more so. If you have an NVIDIA G-Sync compliant GeForce card, does it make sense for

HOW WE TESTED

Initial setup: All monitors to be tested were first examined and everything about them was catalogued, including model number, bundled accessories, available ports, etc. We then reset the displays to factory settings and set brightness, contrast, sharpness at 50%. Depending on the available sync on the monitor two different rigs were set up: one with AMD and the other NVIDIA GPU, respectively. Drivers and essential software were installed. Finally, all performance inhibitors are turned off, as much as possible – stuff like auto screen switching off, auto screen dimming, etc. After this, the monitors are ready to go into the grinder.

Performance tests: We conduct a whole series of tests on the devices to measure the impact of every aspect of the monitor settings on overall performance – tests like brightness, contrast at different gamma and OSD settings. It's essential to calibrate all the displays before subjectively testing them. Therefore, these performance tests are done using the DataColor Spyder 4 Elite test unit – a color calibrator. In dark rooms with little or no ambient light, the monitors are tested with DataColor Spyder Studio. Brightness and contrast is tested at different luminance level, and an average of multiple readings is taken as the final score. We also test the screen's black level and white saturation through Lagom.nl. The performance of the sync objectively tested by running various gaming benchmarks like Grand Theft Auto V and Tomb Raider at the highest refresh rate available on the device. For getting more experiential data we ask some players to play these games and rate it on the basis of how good their experience was, which is highly subjective.

A series of subjective testing is also conducted that scores aspects such as a color reproduction while watching movies, screen readability, onboard speakers, look and feel, playing games (if the monitor is meant for gaming), among others. These tests help complete the gamut of objective benchmarks with real-world test

scenarios, allowing us to take a fairly homogeneous look at the monitor's overall pros and cons.

Performance Scores: After completing all performance tests the final performance score is arrived at, which includes the weighted sum of all the scores catalogued. Weights are assigned based on the perceived importance of the components.

Feature score: The features present in each monitor were tabulated and given weighted scores depending on perceived importance. For example, number of ports on the monitor and screen resolution were given more weightage than panel type. An important note on features, we don't rate sync characteristics any weight because we feel those characteristics already surface in the performance part of our testing.

Design and Usability: The scores for this parameter are largely subjective as they indicate our experience with the device. Items such as bezel thickness, viewing angles, etc. are weighted and scored.

Final score: The final score that we arrive at is the sum of the performance, features and design scores with a variable individual weightage. This is not a fixed weightage, and depending on the category of monitors we're testing, these weightages can be different. For this particular comparison, we gave performance, features and design 50%, 25% and 25% respectively.

A note on winners: The Best Buy monitor is awarded to the device with the best price to performance ratio based on a predetermined formula. This is the ideal, value for money product in the test. The Best Performer award is awarded to the device with the highest overall performance score. The Editor's Pick award is awarded to the device that the reviewer feels is the best product that offers something extra or unique that none of the other monitors have. The Editor's Pick is an entirely subjective award, the decision for which is based on the reviewer's experience while testing the products.

you to go hunting for a G-Sync enabled monitor or not? If you have an AMD FreeSync compliant Radeon GPU, should you just close your eyes and pick a FreeSync enabled display? These technologies are the latest kids on the block, working towards upping your gaming experience.

Apart from this you need a monitor with optimum refresh rate and sync support to render

the in-game frame differences. Believe it or not, the difference between a good monitor and a great monitor is very much visible. Remember playing GTA V, NFS, Battlefield 4 or Just Cause 3? Whenever there was a high speed chase or a building collapsing near your character, you might have noticed frame-lagging, chopping or tearing. These anomalies are efficiently dealt with monitors that have

higher refresh rates and display sync. So a gaming monitor is justified for gaming, on which having sync is an added bonus. But before we start delving into the pros and cons of our tested models let's list all the feature and design aspects you should consider when looking for a gaming monitor:

1. Ports and interfaces: Having many ports and interfaces for connectivity is never a